SeaGlide Project Report

MAE 190 - Mark Anderson

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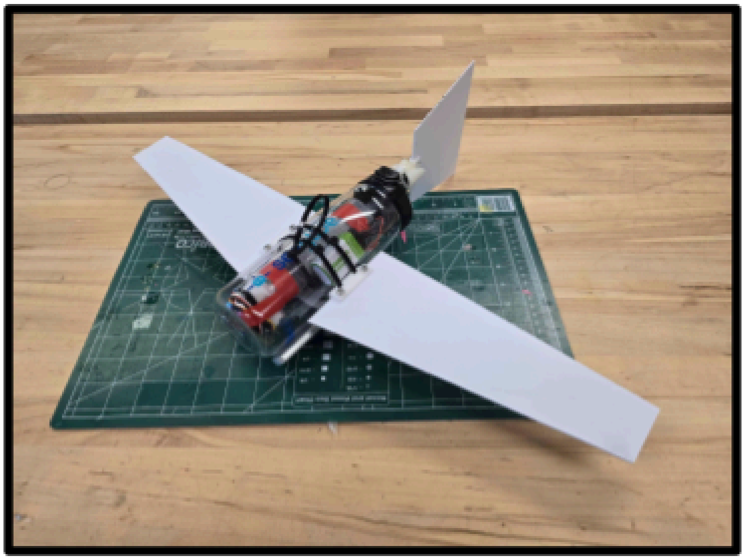
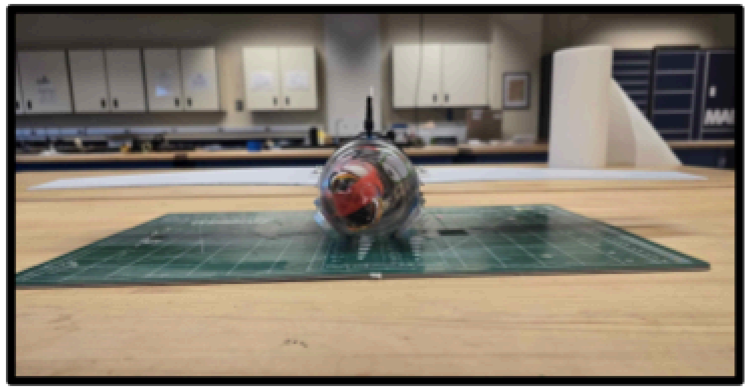
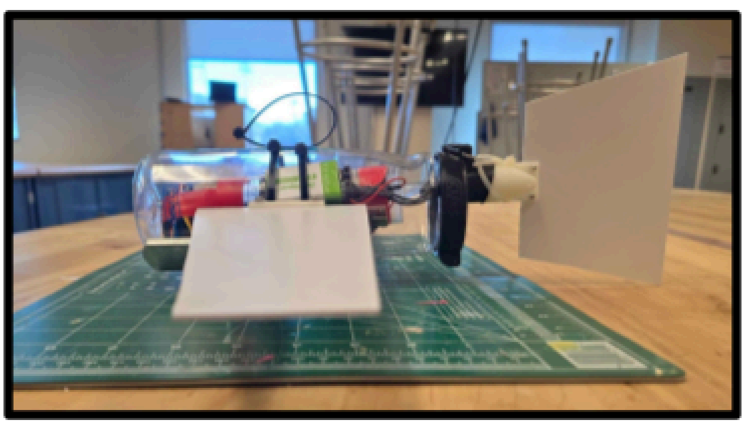
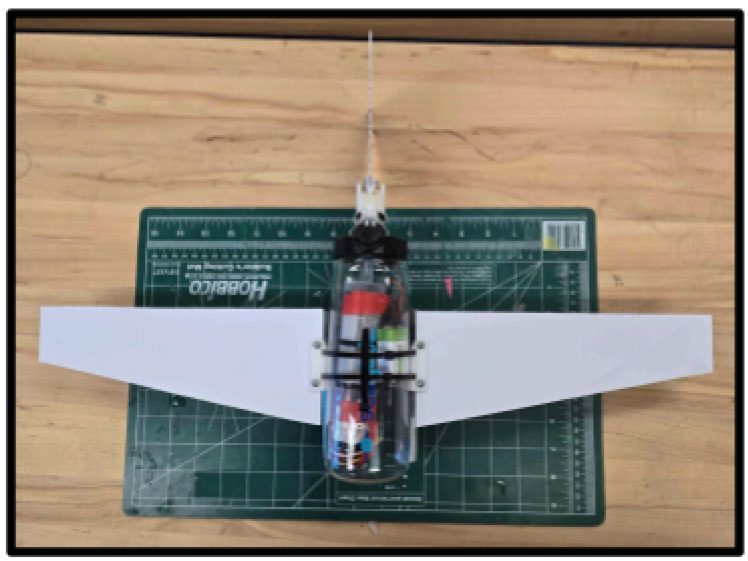








1. Geometry and Dimensions

Overall dimensions

Overall Dimensions	
Wing Span (b)	$51\ cm$
Wing Area (S)	$382.5\ cm^2$
Wing Aspect Ration (AR)	6.8
Dry Weight (Before Balancing)	$9.79\ N$
Dry Weight (After Balancing)	$10.32\ N$
Syringe Volume (Aft/Climb Position)	$\sim 8\ mL$
Syringe Volume (Forward/Dive Position)	$\sim 36\ ml$

2. Buoyancy Engine

The center of gravity (CG) shift is measured relative to the neutral buoyancy CB (which corresponds to the neutral buoyancy CG). In the forward/dive position the CG shift was measured to be $\sim 1.25\ cm$ while in the aft/climb position the CG shift was measured to be $\sim -1.25\ cm$.

Center of Gravity (CG) Shift: $x_G - x_B$	
Forward/Dive Position	$1.25\ cm$
Aft/Climb Position	$-1.25\ cm$

The pitch attitude angle range for equilibrium can be estimated by the CG shift results calculated above. Major assumptions for this estimation include: CG is located approximately 0.5 in (1.27 cm) below CB due to the battery and syringe sitting on the bottom of the water bottle and roll attitude angle $(\phi) = 0$ in equilibrium. Utilizing the equation for equilibrium condition for the moment about the y-axis (local body coordinated): $M_y = 0 = -(z_G - z_B) * \sin\theta - (x_G - x_B) * \cos\theta * \sin\phi$, we obtain the following equation for θ: $\theta = \tan^{-1}\left(\frac{-(x_G - x_B)}{(z_G - z_B)}\right)$

Pitch Attitude Angle (θ) Range	
Forward/Dive Position	44.545°
Aft/Climb Position	-44.545°

Pitch attitude angle range: $\theta = \pm 44.545^\circ$

The buoyancy change can be determined by the weight of the water that is expelled or ingested by the device in the forward/dive position or aft/climb position relative to neutral buoyancy. Since neutral buoyancy was when the the plunger was halfway, either position either expels or ingested 14 mL or water.

Using the equation: $W_{water} = \rho * g * (V_{pos} - V_{neutral})$

Buoyancy Change $(W - B)$ Results	
Forward/Dive Position	$+0.0589\ N$
Aft/Climb Position	$-0.2551\ N$

3. Hydrodynamics

a. Use the wing aspect ratio (AR) to estimate the induced drag coefficient (K) assuming an aerodynamic efficiency of e = 0.8. A parasitic drag coefficient of CD0 = 0.10 can be assumed as well.

$$K = \frac{1}{\pi e(AR)} = \frac{1}{\pi(0.8)(6)} = 0.066$$

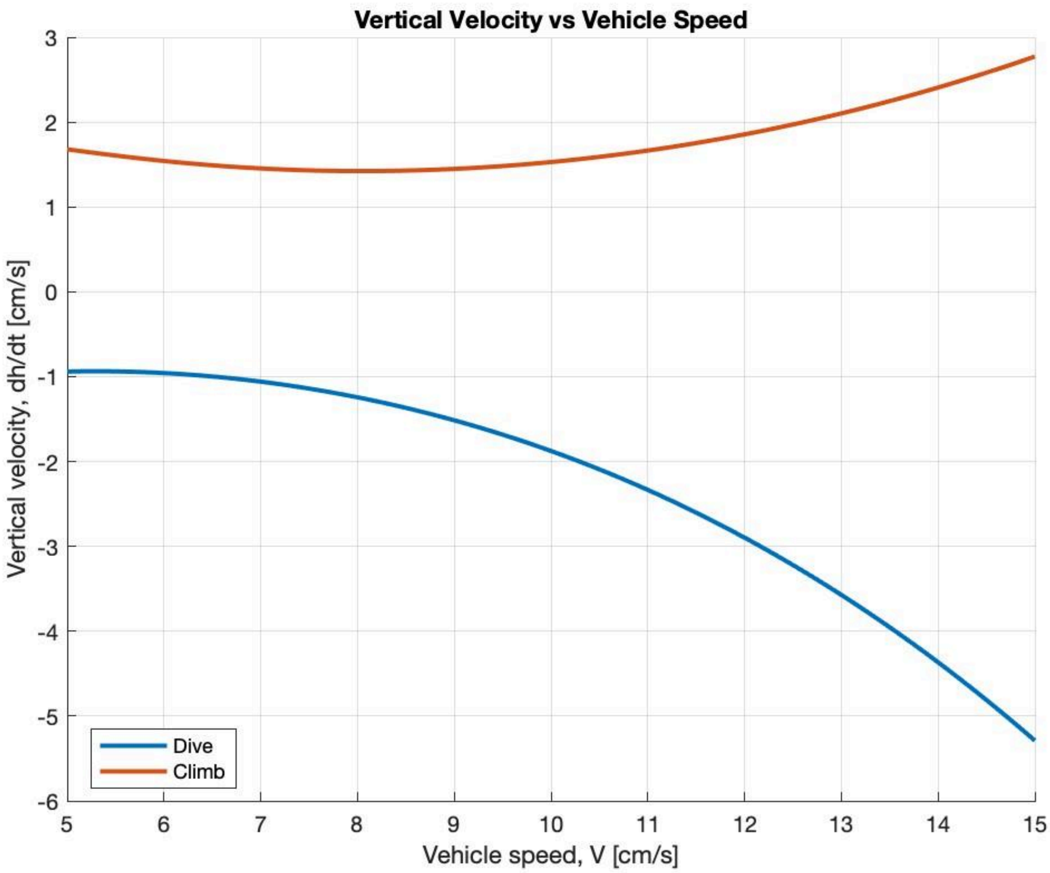
b. Use the wing aspect ratio (AR) to estimate the vehicle lift-curve- slope (CL_alpha) assuming a NACA 0012 airfoil.

Using $c_{l,\alpha} = \frac{\Delta c_l}{\Delta \alpha_0} = \frac{1}{8^\circ * \frac{\pi}{180^\circ}} = 7.162[1/rad]$

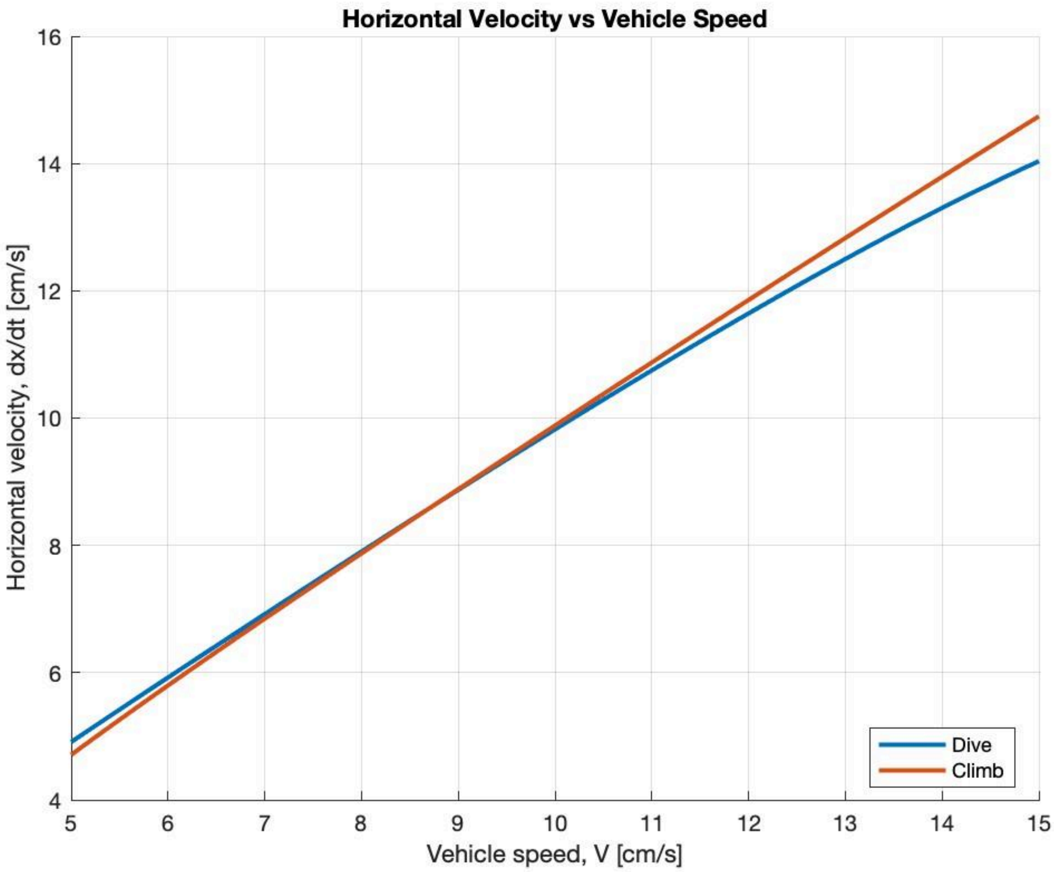
$$C_{L,\alpha} = c_{l,\alpha}\left(\frac{AR}{AR+\left(\frac{2(AR+4)}{AR+2}\right)}\right) = 7.162\left(\frac{6.8}{6.8+\frac{2*10.8}{8.8}}\right) = 5.2625$$

4. Performance Prediction Plots

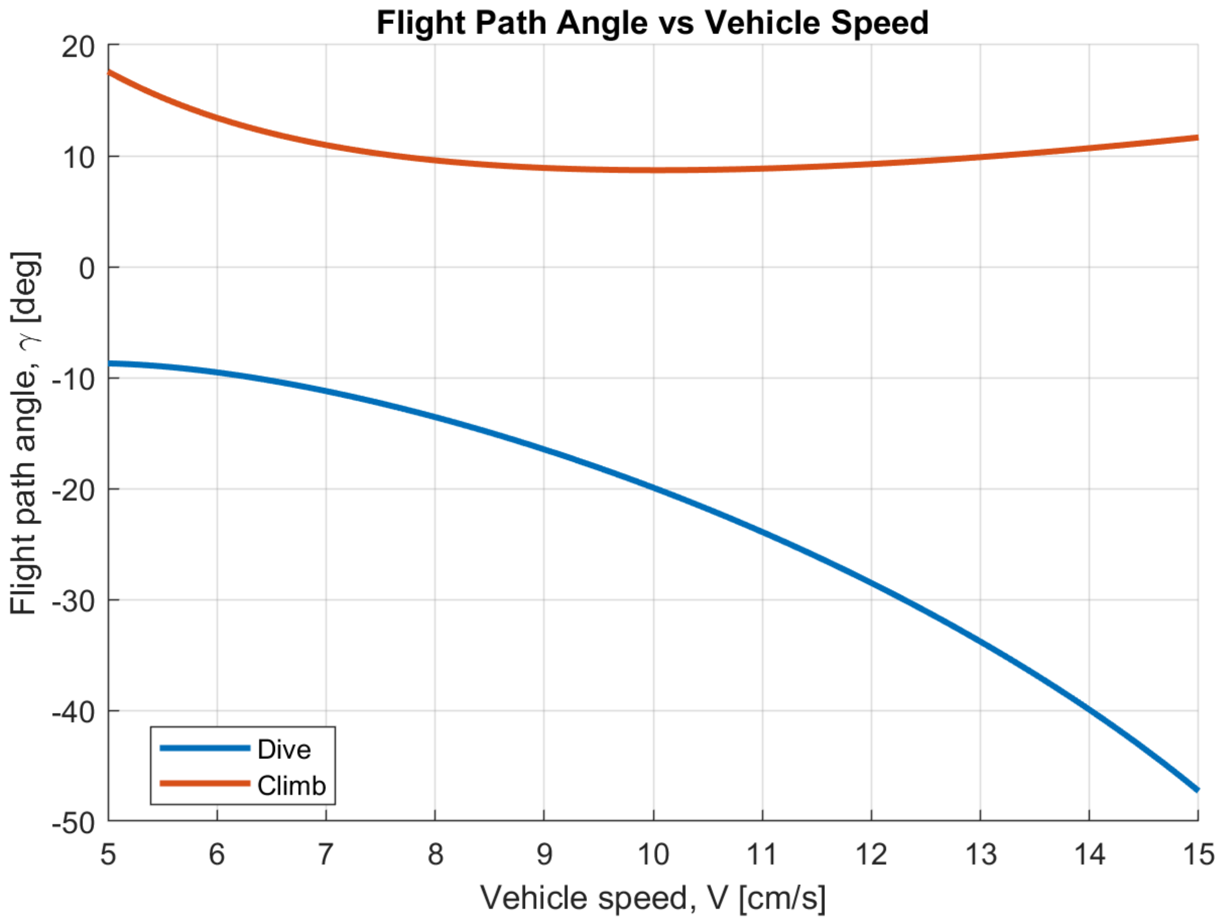
a. vertical velocity, dh/dt [units: cm/s]



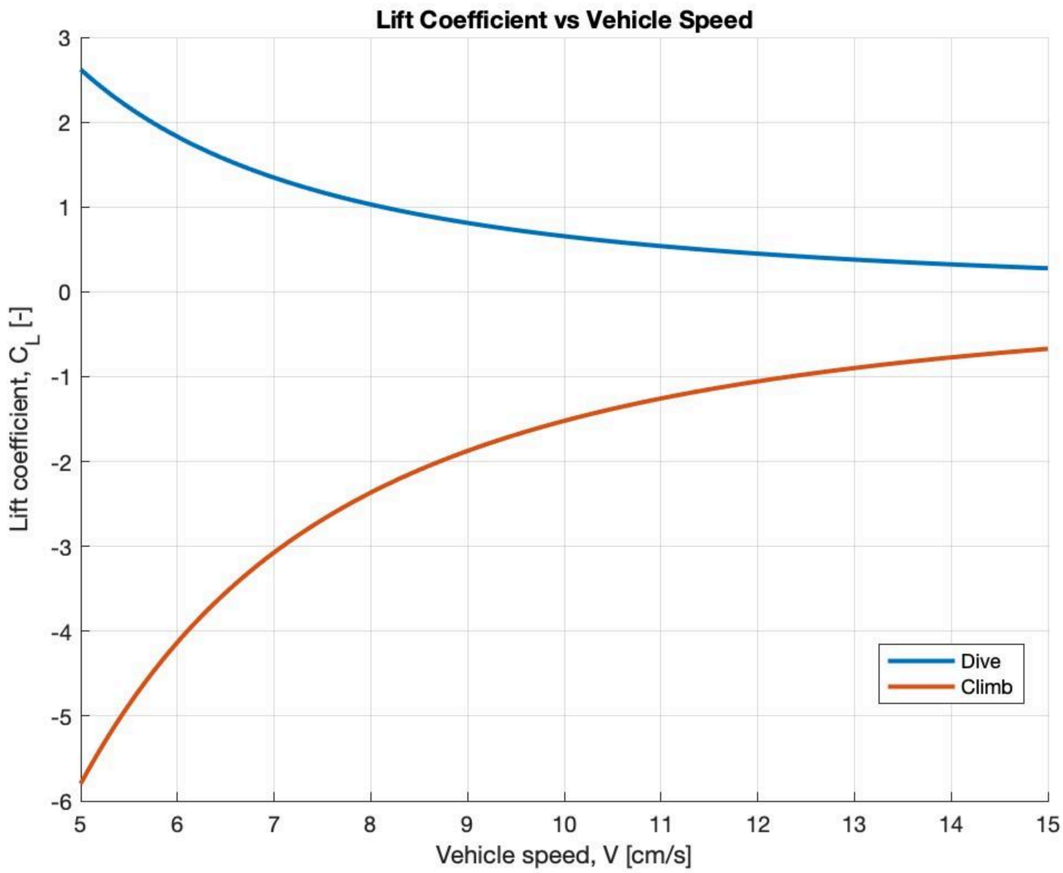
b. horizontal velocity, dx/dt [units: cm/s]



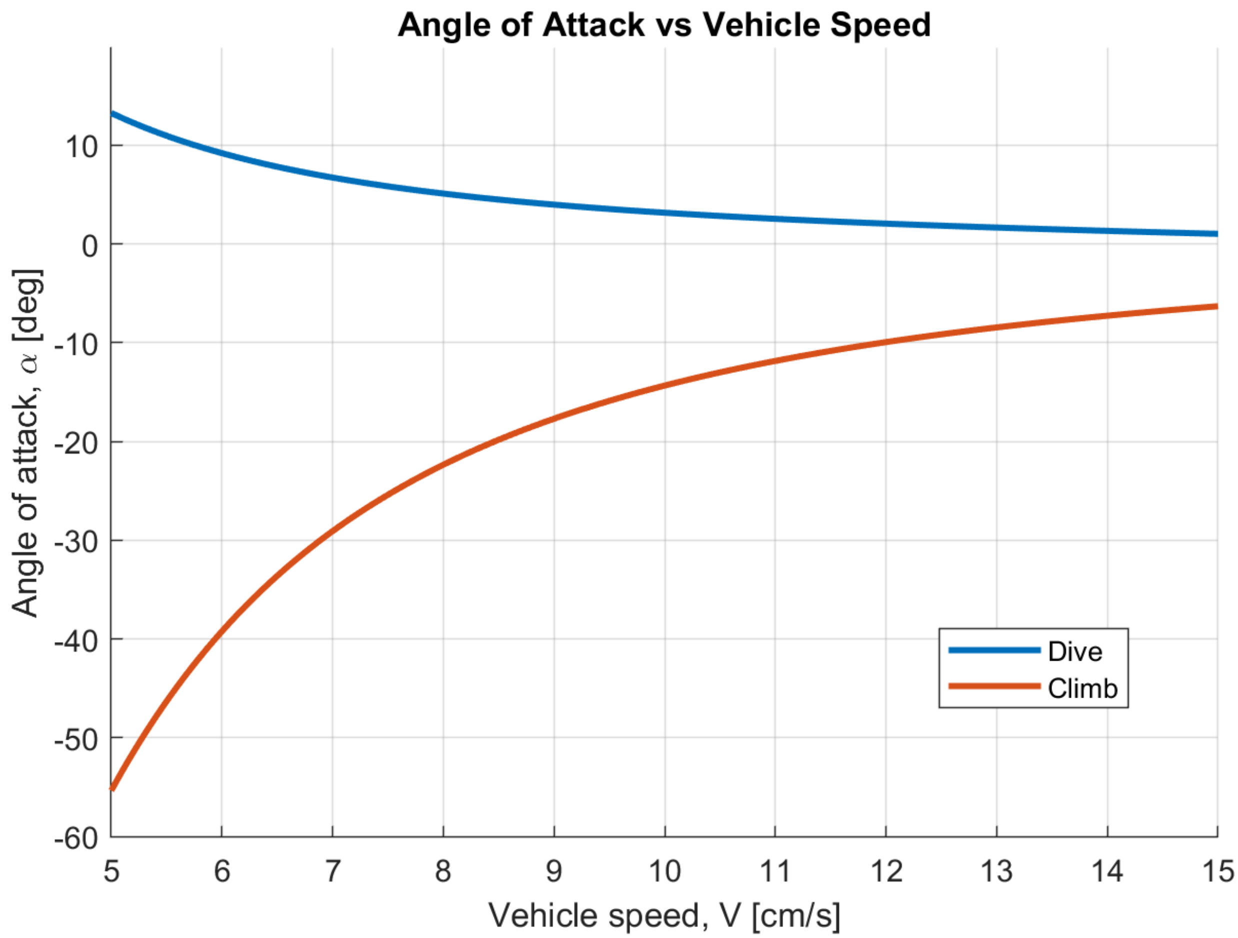
c. flight path angle, gamma [units: deg]



d. lift coefficient, CL



e. angle-of-attack, alpha [units: deg]



f. pitch attitude angle, theta [units: deg]